

Karthikheyaa Kurra

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EDUCATION

University of Delaware

M.S. Data Science, GPA 3.80/4.00

Newark, DE

Aug 2024 – May 2026

- **Coursework:** Probability Theory & Applications, Mathematical Techniques for Data Science (Optimization), Applied Multivariate Statistics, Machine Learning, Algorithm Design & Analysis, Research Software Engineering

Sathyabama Institute of Science & Technology

B.E. Computer Science & Engineering

Chennai, India

Oct 2020 – May 2024

EXPERIENCE

Research Associate, Scientific Machine Learning

Jun 2025 – Present

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Developing hybrid classical and neural solvers for semilinear elliptic PDEs: learned surrogates that replace iterative Newton solves with a single forward pass, amortizing solution cost across problem instances (manuscript in preparation).
- Built the project's end-to-end research infrastructure: high-precision ground-truth generation via Chebyshev spectral discretizations and Newton iteration, automated dataset-validation suites, and reproducible PyTorch training and evaluation pipelines.
- Drove architecture and loss design through controlled ablation studies covering network normalization and physics-informed residual losses with adaptive loss balancing, alongside systematic wall-clock profiling of classical versus learned inference.
- Maintain a regression-locked, unit-tested codebase with automated experiment reports that set weekly research direction with the PI.

Research Software Engineer, LLM Serving & Benchmarking

Jun 2026 – Present

Dept. of Computer & Information Sciences, UD (Advisor: Dr. Sunita Chandrasekaran)

Newark, DE

- Member of a research software engineering team building an on-premise LLM assistant for University of Delaware students, faculty, and staff.
- Benchmarking LLM inference and serving performance (throughput, latency, model selection) on NVIDIA DGX hardware to inform deployment decisions.

Research Intern, Computer Vision

Mar 2023 – Jun 2023

Indian Institute of Technology, Hyderabad

Hyderabad, India

- Case study applying supervised contrastive learning (Khosla et al., 2020) to facial expression recognition, comparing SupCon loss against a standard cross-entropy baseline in PyTorch with face detection and data augmentation.
- Analyzed learned embeddings with t-SNE to diagnose class separation issues and delivered a reproducible codebase to the lab.

Teaching Assistant, Machine Learning

Jan 2023 – Apr 2023

Sathyabama Institute of Science & Technology

Chennai, India

TECHNICAL SKILLS

Languages: Python, MATLAB, R, SQL, Bash, LaTeX

Scientific ML: PyTorch, NumPy/SciPy, Operator Learning, Physics-Informed Neural Networks (PINNs)

Tools: Git, Docker, Linux, HPC job scripting, VS Code, Cursor

READING GROUPS

Scientific Machine Learning Reading Group

2025 – Present

University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Weekly seminar with rotating student presentations of research literature; topics include Computational Optimal Transport (Peyré & Cuturi), Equivariant Neural Networks, and One-Shot Learning.